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1. A curve has equation

$$y = 2 - \frac{3}{x+1}$$

(a) State the equations of the horizontal and vertical asymptotes of this curve. [2]

(b) The straight line $y = 3x + c$ meets the curve. Show that the x -coordinates of the points of intersection satisfy

$$3x^2 + (c+1)x + (c+1) = 0$$
 [3]

(c) It is given that the line $y = 3x + c$ is a tangent to the curve.

(i) Without using calculus, find the two possible values of c . [3]

(ii) Hence find the coordinates of each point of contact. [4]

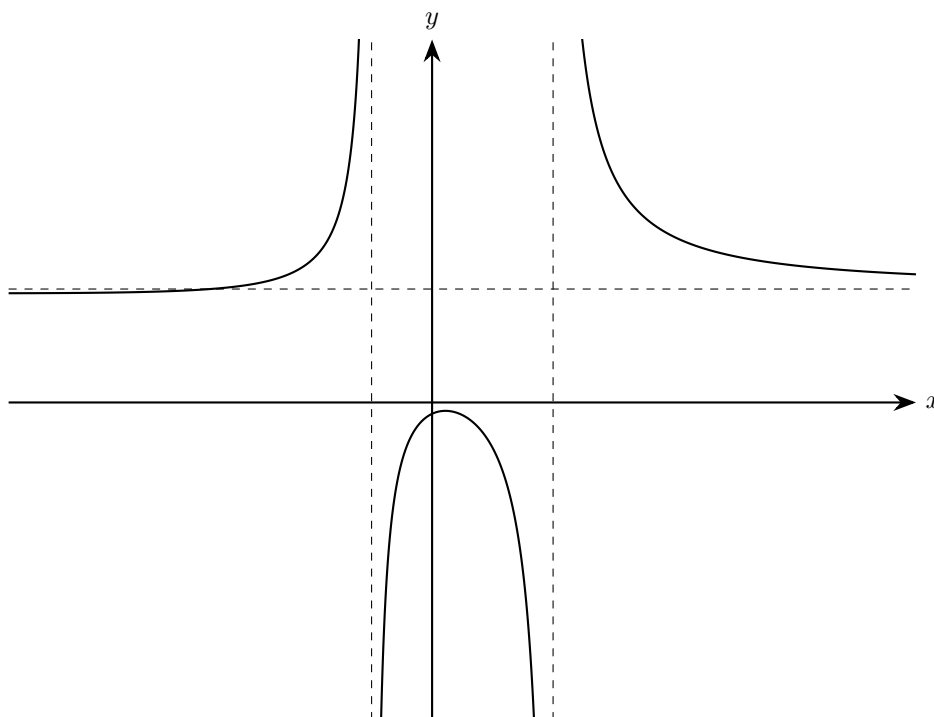
2. The curve C has equation $y = \frac{x+1}{x^2-4x+3}$.

(a) Find the equations of the asymptotes of C . [2]

(b) Find the coordinates of any stationary points on C . [4]

(c) Sketch C , stating the coordinates of the intersections with the axes. [3]

(d) Sketch the curve with equation $y = \left| \frac{x+1}{x^2-4x+3} \right|$ and find in exact form the set of values of x for which $2|x+1| > |x^2-4x+3|$. [6]



3. The diagram shows the curve C with equation

$$y = \frac{5x^2 + ax - b}{x^2 + bx + a}$$

where a and b are integers.

The vertical asymptotes of C are $x = -1$ and $x = 2$.

- (a) State the equation of the asymptote of C that is parallel to the x -axis. [1]
- (b) Use the vertical asymptotes to find the value of a and the value of b . [2]
- (c) Hence, or otherwise, find the coordinates of the point where C meets the y -axis. [1]
- (d) Without using calculus, show that the line $y = 4$ does not intersect C . [5]

4. The function f is defined by

$$f(x) = \frac{-2x^2 + 11x - 20}{2x - 3} \quad (x \in \mathbb{R}, x \neq \tfrac{3}{2})$$

(a) Find the range of f [5]

(b) Find the coordinates of the two stationary points of the graph of $y = f(x)$. [2]

(c) Show that the graph of $y = f(x)$ has an oblique asymptote and find its equation. [2]

(d) Sketch the graph of $y = f(x)$, showing clearly the asymptotes and stationary points. [4]

5. The curve C has equation $y = \frac{x^2 + ax - 6}{x + 1}$, where $a > 0$.

(a) Find the equations of the asymptotes of C . [3]

(b) Show that C has no stationary points. [4]

(c) Sketch C , stating the coordinates of the point of intersection with the y -axis and labelling the asymptotes. [3]

(d) (i) Sketch the curve with equation $y = \left| \frac{x^2 + ax - 6}{x + 1} \right|$. [2]

(ii) On your sketch in part (i), draw the line $y = a + 1$. [1]

(iii) It is given that $\left| \frac{x^2 + ax - 6}{x + 1} \right| < a + 1$ for $-11 < x < -3$ and $0 < x < 4$.
Find the value of a . [2]

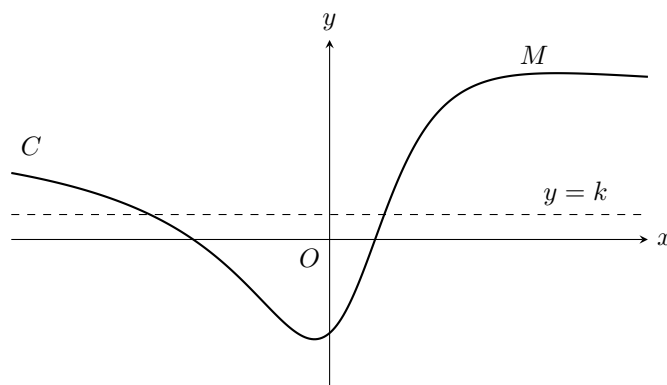
6. The function f is defined by

$$f(x) = \frac{x^2 + 2x - 3}{x^2 + px + 4} \quad x \in \mathbb{R}$$

where p is a constant.

The graph of $y = f(x)$ has only one asymptote.

- (a) Write down the equation of the asymptote. [1]
- (b) Find the set of possible values of p . [4]
- (c) Find the coordinates of the points at which the graph of $y = f(x)$ intersects the coordinate axes. [3]



(d) A curve C has equation

$$y = \frac{x^2 + 2x - 3}{x^2 - x + 4}$$

The curve C has a local maximum at the point M as shown in the diagram.
The line $y = k$ intersects curve C .

(i) Show that

$$15k^2 - 8k - 16 \leq 0 \quad [5]$$

(ii) Hence, find the y -coordinate of point M . [2]

7. A curve has equation

$$y = \frac{x^2 - 9}{x^2 - 4}$$

(a) Write down the equations of the asymptotes to the curve. [3]

(b) Sketch the curve, indicating the coordinates of the points where the curve intersects the coordinate axes. [5]

(c) Hence, or otherwise, solve the inequality

$$0 < \frac{x^2 - 9}{x^2 - 4} < 1 \quad [2]$$

8. The curve C has equation

$$y = \frac{x^2 + 3}{x + 1}$$

(a) Find the equations of the asymptotes of C . [3]

(b) Show that there is no point on C for which $-6 < y < 2$. [4]

(c) Sketch C . [2]

(d) (i) Sketch the graphs of

$$y = \left| \frac{x^2 + 3}{x + 1} \right| \text{ and } y = |x - 1|$$

on the same axes, stating the coordinates of any intersections with the axes. [4]

(ii) Use your sketch to find the set of values of c for which

$$\left| \frac{x^2 + 3}{x + 1} \right| \leq |x - 1| + c$$

has no solution. [1]

9. Let a be a positive constant.

(a) The curve C_1 has equation

$$y = \frac{x^2 - a^2}{x^2 - 4a^2}$$

Sketch C_1 .

[2]

The curve C_2 has equation

$$y = \left(\frac{x^2 - a^2}{x^2 - 4a^2} \right)^2$$

The curve C_3 has equation

$$y = \left| \frac{x^2 - a^2}{x^2 - 4a^2} \right|$$

(b) (i) Find the coordinates of all stationary points of C_2 .

[3]

(ii) Find also the coordinates of all points of intersection of C_2 and C_3 .

[3]

(c) Sketch C_2 and C_3 on a single diagram, clearly identifying each curve. Hence find the set of values of x for which

$$\left(\frac{x^2 - a^2}{x^2 - 4a^2} \right)^2 \leq \left| \frac{x^2 - a^2}{x^2 - 4a^2} \right|$$

[5]

10. A curve C_1 has equation

$$y = -2 + \frac{12}{x+3}$$

(a) Write down the equations of the asymptotes of curve C_1 . [2]

(b) Sketch the graph of curve C_1 . Indicate the values of the intercepts of the curve with the axes. [3]

(c) Hence, or otherwise, solve the inequality

$$-2 + \frac{12}{x+3} > 0$$
 [2]

(d) Curve C_2 is a reflection of curve C_1 in the line $y = -x$. Find an equation for curve C_2 in the form $y = f(x)$. [3]

11. The curve C has equation

$$y = \frac{x^2 - 6x + 5}{x + 2}$$

(a) Find the equations of the asymptotes of C . [3]

(b) Find the exact coordinates of the stationary points on C . [4]

(c) Sketch C , stating the coordinates of any intersections with the axes. [3]

(d) Sketch the curve with equation

$$y = \left| \frac{x^2 - 6x + 5}{x + 2} \right|$$

and find in exact form the set of values of x for which

$$\left| \frac{x^2 - 6x + 5}{x + 2} \right| < 6 \quad [5]$$

12. A curve has equation $y = \frac{x-1}{x^2-x-6}$.

- (a) (i) Write down the equations of the three asymptotes of the curve. [3]
(ii) Sketch the curve, showing the coordinates of any points of intersection with the coordinate axes. [4]

(b) Hence, or otherwise, solve the inequality

$$\frac{x-1}{x^2-x-6} > \frac{1}{2} \quad [3]$$

13. The curve C has equation

$$y = \frac{(x-1)(x+3)}{x^2 - 2x + 3}$$

(a) Show that C has no vertical asymptotes and state the equation of the horizontal asymptote of C . [3]

(b) Find the coordinates of the stationary points on C . [4]

(c) Sketch C , stating the coordinates of the intersections with the axes. [3]

(d) Sketch the curve with equation

$$y = \left| \frac{(x-1)(x+3)}{x^2 - 2x + 3} \right|$$

and state the set of values of k for which

$$\left| \frac{(x-1)(x+3)}{x^2 - 2x + 3} \right| = k$$

has 4 distinct real solutions.

[2]

14. The curve C has equation

$$y = \frac{-x^2 + 4x + 5}{x + 2}$$

(a) Find the equations of the asymptotes of C . [3]

(b) Find the coordinates of the stationary points of C . [4]

(c) Sketch C , stating the coordinates of the intersections with the axes. [3]

(d) Sketch the curve with equation

$$y = \left| \frac{-x^2 + 4x + 5}{x + 2} \right|$$
 [2]

(e) Using your sketch, or otherwise, find the set of values of x for which

$$\left| \frac{-x^2 + 4x + 5}{x + 2} \right| < 2$$
 [4]